SMART WASTE SORTING BIN

Detailed Design Package · Version 2.0

Standalone edge AI · ESP32-S3-CAM + MobileNetV2 + sensor fusion

420 × 420 × 900 mm · 4 × 8.7 L · BOM HK$866 · total ≈ HK$996

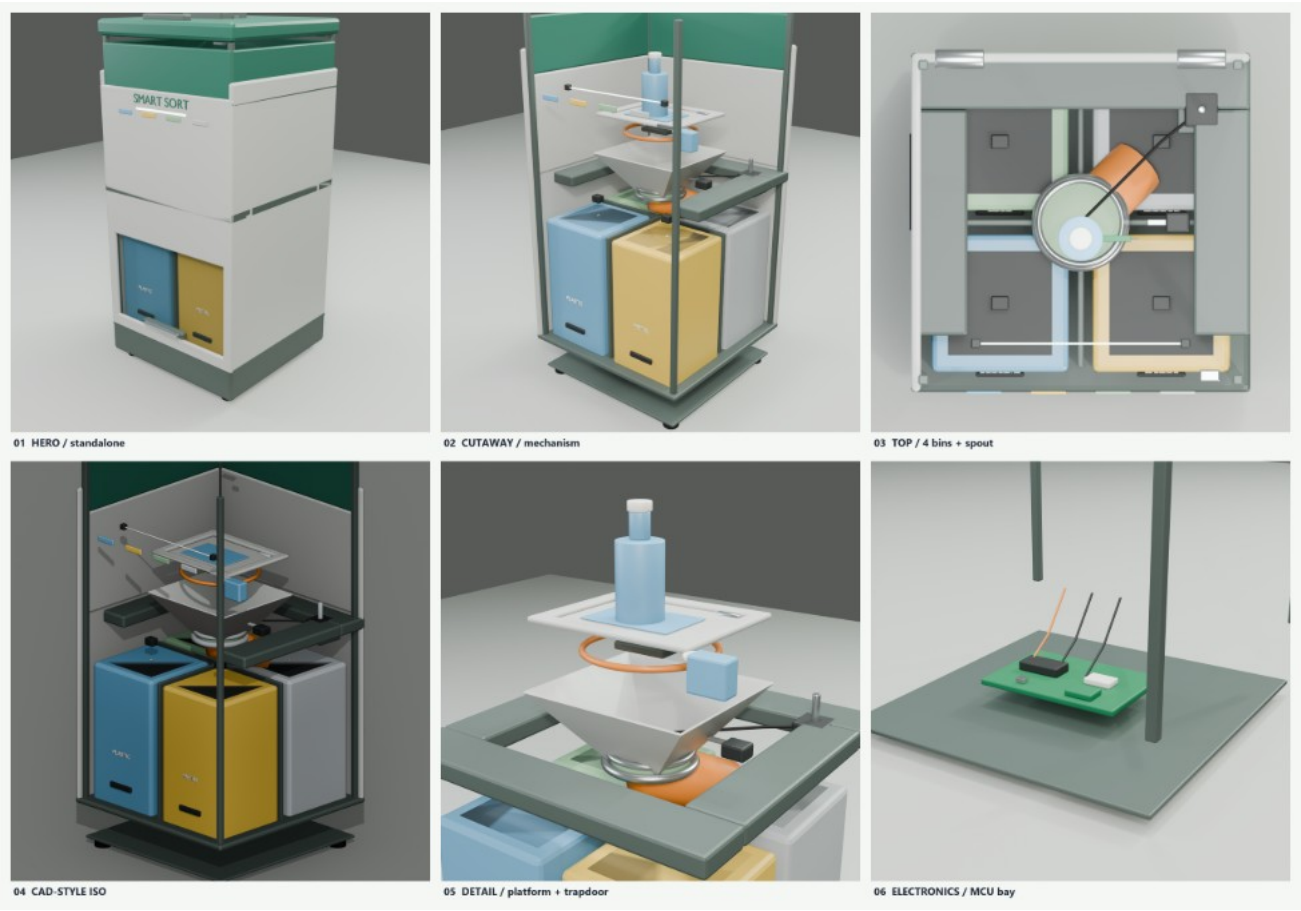


Figure 0. Current v2 model: exterior, cutaway, top view, CAD style, mechanism detail and electronics bay.

No laptop · no cloud · no agent. The ESP32-S3-CAM performs inference on-board; WiFi AP and OLED are for status only.

Detailed version, separate from the 9-page concise v2 PDF. Supersedes the archived HK$4,050 document for the current budget.

Contents

1. 1. Project definition and requirements
2. 2. Design decisions and architecture
3. 3. Mechanical design
4. 4. Electrical design and pin map
5. 5. Firmware, AI and sensor fusion
6. 6. Cost and bill of materials
7. 7. Build, calibration and test plan
8. 8. Risks, limitations and future work

This detailed report is written to be buildable: it gives dimensions, cut lists, pin assignments, state logic, calibration steps, test criteria and a costed BOM. It is still a concept/visualisation design, not a manufacturing drawing.

1. Project definition and requirements

1.1 Problem and objective

Hong Kong's recycling stream is contaminated by items placed in the wrong bin. A four-stream bin that classifies one item at a time, records the decision and physically moves the item into the correct bin demonstrates a complete mechatronic loop: sense, decide, actuate, verify. The unit must be standalone and cost no more than HK$1,000.

The engineering objective is not a commercial product. It is a final-year prototype that can be built with hand tools, standard bins and a small number of purchased modules, then evaluated on a 50-item mixed dataset.

1.2 Scope and operating envelope

Item	Design value	Reason
Streams	Plastic bottle, metal, wet, dry/other	Four visible bins
Item size	≤ 500 mL bottle; ≤ 0.5 kg	Trapdoor / servo torque
Operation	One item at a time; lid closed	Camera and weighing repeatability
Environment	Indoor, 5–40 °C, dry floor	Electronics not weatherproof
Power	12 V 3 A adapter; 5 V buck	No Li-ion pack
Users	One user; phone optional	WiFi AP is not required to sort

Out of scope: bags, multiple items, hazardous waste, glass, liquids, 2 L bottles, heavy bags, outdoor use, and any cloud training at run time.

1.3 Functional and non-functional requirements

ID	Requirement	Target
FR1	Classify one item into 4 streams	≥80–85% on 50 items
FR2	Weigh the item	0–5 kg; ±10 g
FR3	Detect wet / metal evidence	Moisture threshold; metal confidence
FR4	Move spout to selected bin	4 positions; ±3°
FR5	Open trapdoor and drop item	≤0.5 kg; no jam in 20 cycles
FR6	Show state / class on OLED	128×64, I2C
FR7	Log to microSD	CSV: class, weight, time
NFR1	Standalone; no laptop	All inference on-board
NFR2	Safe low voltage	12 V SELV; fuse; e-stop
NFR3	Cost	BOM ≤ HK$1,000; target ≤996
NFR4	Buildable with hand tools	±2 mm tolerance; no welding
NFR5	Privacy	Images stay on device

1.4 Use cases and stakeholders

Actor	Scenario	Expected behaviour
Primary user	Throw one bottle, close lid	Classify, weigh, drop, show result
Primary user	Unsure item	Low-confidence path; OLED asks; no wrong drop
Cleaner	Empty bins / clean tray	Drawer and drip tray removable; lockout
Technician	Calibrate / update model	USB-C flash; SD card; serial log
Evaluator	50-item test	CSV + confusion matrix + cycle time
Demo audience	One cycle	Visible motion; no laptop; safe

Assumptions: one item at a time; indoor; 5–40 °C; dry floor; the user can lift the lid; the phone is optional. The machine is not a commercial waste sorter and is not rated for hazardous or medical waste.

2. Design decisions and architecture

2.1 Why standalone ESP32-S3-CAM

The earlier laptop-tethered version was cheap but not a standalone IoT product. The classic ESP32-CAM has only 4 MB PSRAM and is not practical for MobileNetV2. The ESP32-S3-CAM used here has 8 MB PSRAM and 16 MB flash, a USB-C port for flashing/power, a camera connector and microSD. It runs MobileNetV2 0.35/0.5 at 96–128 px, INT8, with TensorFlow Lite Micro.

This is a deliberate compromise: full MobileNetV2 1.0 at 224 px has ~300 M MACs and a ~3.5 MB INT8 model; latency and arena size are not comfortable on this board. The 0.35/0.5 backbone is still MobileNetV2, and transfer learning on four classes is enough for a prototype. The report must state the width multiplier, input size and measured accuracy.

2.2 Key decisions

Decision	Selected	Why
Compute	ESP32-S3-CAM, 8 MB PSRAM	Standalone; no laptop
Spout drive	NEMA 17 + A4988 + 3:1 GT2	1.35 N·m at spout; SF≈4
Weighing	Fixed platform + 5 kg cell + trapdoor	No side-load on cell
Door	MG996R + spring	0.98 N·m; item ≤0.5 kg
Bins	4 × 170×170×300 mm; no notches	Standard bins; spout above
Power	12 V 3 A + 5 V buck	No Li-ion / BMS
UI	OLED + WiFi AP	Phone optional; no cloud

2.3 System architecture

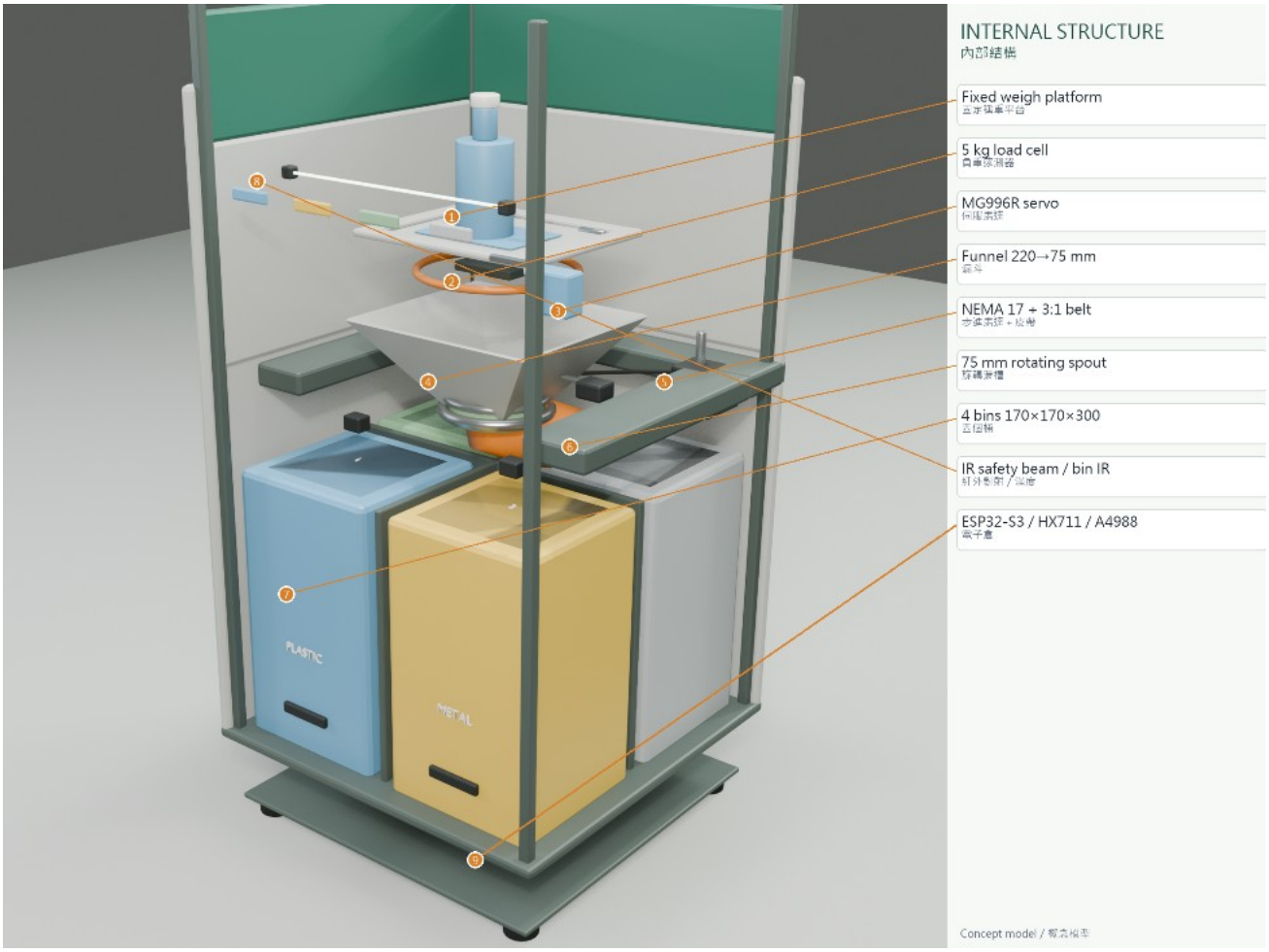


Figure 1. Internal structure and component locations.

Five layers: (1) physical: lid, platform, trapdoor, funnel, spout, four bins; (2) sensing: ESP32-S3-CAM, 5 kg load cell + HX711, moisture pad, IR beam, four bin-full IR, home/lid switches; (3) actuation: NEMA 17 + A4988 + GT2 belt, MG996R trapdoor; (4) compute: ESP32-S3, on-board MobileNetV2, fusion rules, state machine; (5) UI/logging: OLED, WiFi AP, microSD.

The item path is fixed: item → trapdoor → fixed platform → load cell → mid-frame. The spout rotates under the funnel; the selected bin receives the item. There is no rotating bin carousel and no corner notch, which keeps the structure simple and the bins replaceable.

2.4 Alternatives considered

Option	Problem	Decision
Laptop-tethered USB camera	Not standalone; cable	Rejected for this brief
Raspberry Pi Zero 2 W	Linux; more power; more cost	Possible but not ESP32-CAM
Classic ESP32-CAM	4 MB PSRAM; MobileNetV2 too large	Rejected
ESP32-S3-CAM, 8 MB PSRAM	Best fit for on-device model	Selected
Rotating bin carousel	Heavy; large bearing; notches	Rejected
Tilting load-cell tray	Side-load; drift	Rejected

Fixed platform + trapdoor	Repeatable weighing	Selected

2.5 Requirement traceability

Requirement	Design element	Verification
FR1 classify	S3-CAM + MobileNetV2 + fusion	50-item test
FR2 weigh	Fixed platform + 5 kg cell	UT2 0–1 kg
FR3 wet/metal	Moisture pad + vision/weight	UT4 / 50-item
FR4 spout	NEMA 17 + 3:1 + home	UT5 ±3°
FR5 drop	MG996R + trapdoor + spring	UT6 20 cycles
FR6 display	OLED I2C	Visual check
FR7 log	microSD CSV	Export / inspect
NFR1 standalone	On-board inference	No laptop in demo
NFR2 safety	Fuse, e-stop, interlocks	Fault injection

3. Mechanical design

3.1 Overall dimensions and layers

Layer	Height	Contents	Access
Base / electronics	90 mm	ESP32-S3, HX711, A4988, 5 V buck, terminals	Rear panel
Bins	300 mm	4 × 170×170×300 mm bins	Front drawer / lift
Mid-frame	60 mm	Bearing, NEMA 17, spout home, funnel	Top access
Hopper	225 mm clear	Platform, trapdoor, load cell, servo, moisture pad	Lid
Lid	40 mm	ESP32-S3-CAM, lens, LED ring, IR, reed	Rear hinge

Overall 420 W × 420 D × 900 H mm. Tray top 635 mm; lid underside 860 mm; clear height 225 mm, enough for an upright 500 mL bottle. The top collar is not decorative: it provides the clearance that the first 780 mm sketch lacked.

3.2 Frame, bins and drawer

Use 12 mm plywood or 5 mm acrylic on a 20 × 20 mm pine/aluminium cleat frame. The base is 420 × 420 mm; four 170 × 170 × 300 mm bins sit in a 2 × 2 grid with 30 mm between bins for the spout bearing. A removable drip tray sits under the bins. The front window shows the two front bins; the rear bins are labelled on the top and front.

Bins are not notched. The 75 mm spout rotates above the bin tops and discharges through a 100 mm radius outlet. The outlet must not touch the bin rim; keep 10–15 mm clearance. If the bin is taller than 300 mm, reduce the hopper height or use a shorter bin; do not simply cut the spout.

3.3 Weigh platform and trapdoor

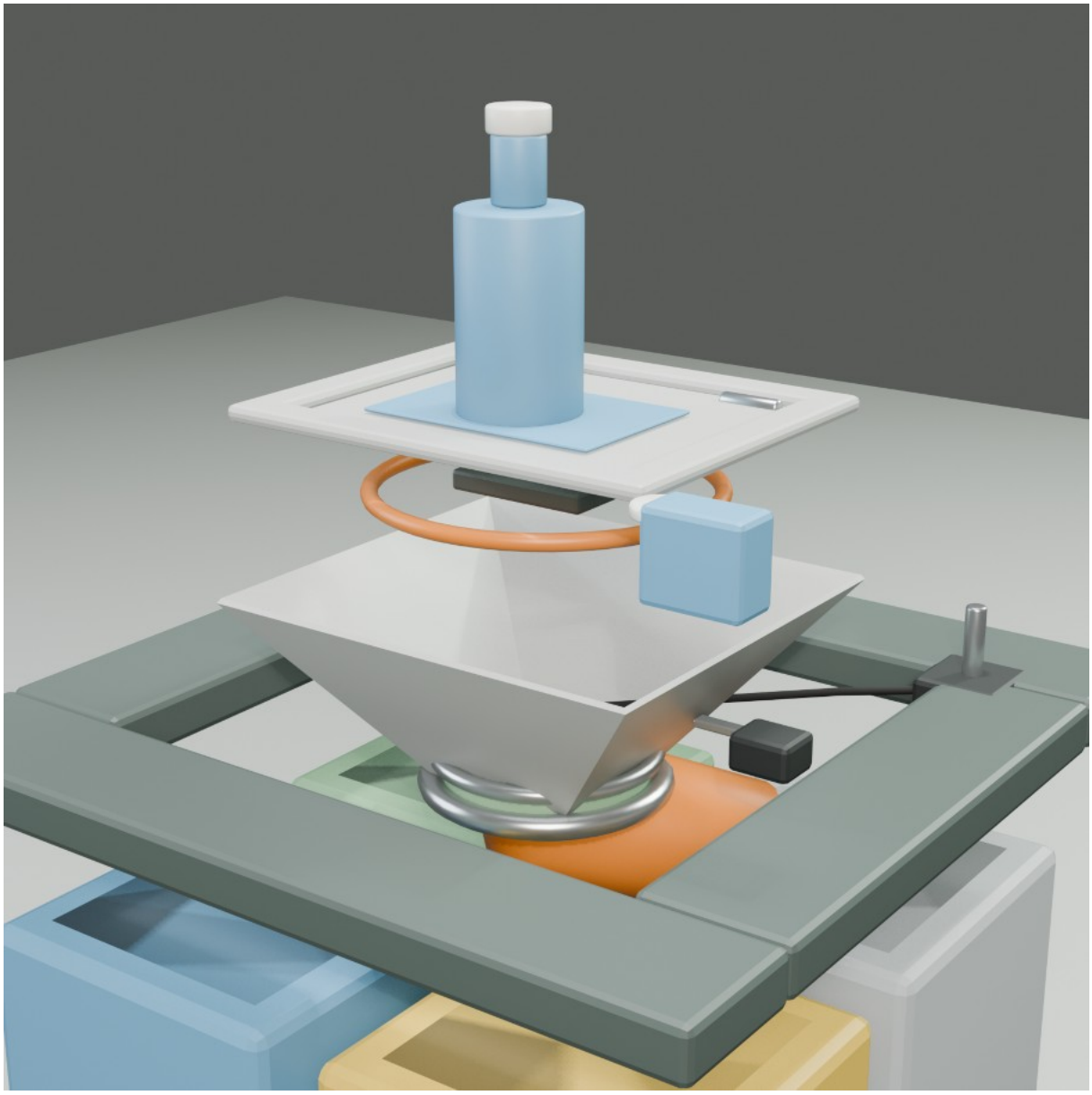


Figure 2. Fixed platform, trapdoor, load cell, metal coil, servo and funnel.

Part	Size / spec	Note
Platform frame	220 × 180 mm, 12 mm ply	Fixed to mid-frame
Trapdoor	180 × 140 mm, 6 mm ply/PE	Hinge at back
Load cell	5 kg single-point	Under platform centre
Servo	MG996R, 10 kg-cm	Side; spring return
Moisture pad	Capacitive, 60 × 30 mm	On door; HDPE cover
Metal coil	Optional 60–100 mm	Fixed under platform

Load path

Item → trapdoor → platform → load cell → mid-frame. The servo and hinge are on the platform frame; the load cell

never carries their side-load. Weigh before opening the door.

3.4 Funnel, bearing and spout

Funnel: 220 × 200 mm top → 75 mm outlet, 90 mm high, 3 mm PP/PETG. A 110 mm lazy-susan bearing carries the spout collar. The spout is a 75 mm lightweight 3D-printed elbow (PETG, 1.6 mm wall) with a 100 mm outlet radius. NEMA 17 drives a 20T motor pulley and 60T spout pulley through a 200 mm GT2 belt; a slotted bracket tensions the belt.

Home switch: a lever microswitch on the mid-frame, actuated by a cam on the spout collar. Homes at 45°; positions at 45°, 135°, 225°, 315°. Always approach clockwise; 1200 steps per 90° at 1/8 microstep and 3:1.

3.5 Torque and clearance checks

Check	Available	Required	Margin
Spout rotation	0.45 × 3 = 1.35 N·m	≈0.34 N·m	≈4.0×
Trapdoor	MG996R 0.98 N·m	≈0.34 N·m	≈2.8×
Load cell	5 kg	≤0.8 kg	≈6×
Bottle clearance	225 mm	≈200 mm	25 mm
Bin-full IR	fixed height	80% warning	threshold only

These are nominal calculations. The spout is 3D-printed and its mass must be measured; if it exceeds 0.45 kg or the bearing friction is high, use a 4:1 or 5:1 belt and re-check speed. The MG996R must be spring-assisted and the item kept below 0.5 kg.

3.6 Cut list and tolerances

Part	Size (mm)	Qty	Material
Base	420 × 420	1	12 mm ply
Side	420 × 900	2	12 mm ply
Back	396 × 900	1	12 mm ply
Top	420 × 420	1	12 mm ply
Mid-frame	420 × 420	1	12 mm ply
Bin dividers	396 × 300	2	12 mm ply
Platform frame	220 × 180	1	12 mm ply
Trapdoor	180 × 140	1	6 mm ply/PE
Front window	300 × 220	1	5 mm acrylic
Drip tray	396 × 396	1	PP sheet

Tolerance: ±2 mm on plywood, ±0.5 mm on 3D-printed bearing collar, ±1 mm on hinge/door gaps. Drill pilot holes; do not force screws into plywood edges. Sand and seal the bin compartment; wet waste is the main corrosion path.

3.7 Assembly sequence

1. Cut the base, sides, back, top and mid-frame; drill pilot holes and test-fit.
2. Assemble the base and bin compartment; fit the four bins and drip tray.
3. Install the mid-frame, bearing and spout collar; check rotation by hand.
4. Mount NEMA 17, pulleys and belt; tension the belt; fit the home switch.
5. Fit the funnel; seal the joints; check that the spout clears the bin rims.
6. Mount the fixed platform, load cell, trapdoor, hinge and MG996R; check the spring return.
7. Fit the lid, ESP32-S3-CAM, lens, LED ring and reed switch; route the camera flex.
8. Wire the base: 12 V input, fuse, e-stop, 5 V buck, ESP32-S3, HX711, A4988, OLED, microSD.
9. Flash the firmware; home the spout; calibrate the load cell and moisture pad.
10. Run 20 dry cycles, then 20 real items; only then fit the outer panels and labels.

3.8 Materials, finish and tolerance stack

Part	Material	Thickness / spec	Finish
Frame	Plywood / acrylic	12 mm / 5 mm	Sand, seal edges
Hopper	PP / PETG	3 mm / 1.6 mm printed	Matte black inside
Spout	PETG	1.6 mm wall	Smooth bore
Door	Plywood / HDPE	6 mm	Food-safe cover
Bins	PP	Reused 5–8 L	Label + liner
Bearings	Steel / POM	110 mm lazy susan	Dry lubricant

- Tolerance stack: frame ±2 mm, bearing collar ±0.5 mm, door gap 2–3 mm, spout-to-bin 10–15 mm. Measure the printed collar before drilling the bearing mounting holes.
- Do not paint the inside of the hopper; paint flakes contaminate the waste stream and confuse the camera. Use matte black sheet or tape if glare is a problem.

3.9 Maintenance and cleaning

Task	Frequency	Method
Empty bins	Daily / when full	Remove liner; wipe
Clean trapdoor / pad	Daily	Damp cloth; dry; no solvent
Clean camera lens	Weekly	Microfibre; no pressure
Check belt tension	Monthly	5 mm deflection; adjust
Check hinge / spring	Monthly	Lubricate; replace worn
Calibrate load cell	Monthly / after move	0/500 g check

3.10 Drawing and figure index

Figure	Content	Use
00_overview	Six views	Report cover / overview

01_hero	Exterior	Product image
02_cutaway	Internal mechanism	Mechanical explanation
03_top	Bin layout	Layout / sensor map
05_detail	Platform / door	Weighing explanation
07_internal_labeled	Labeled cutaway	Viva / report
08_electronics_labeled	MCU bay	Wiring explanation
09_storyboard_5	Five-step sequence	Operation / video

4. Electrical design and pin map

4.1 Power architecture

Rail	Source	Loads	Note
12 V	12 V 3 A adapter	A4988 VMOT, buck	Fuse 3 A; e-stop
5 V	5 V 3 A buck	ESP32-S3, OLED, sensors	1000 µF bulk
5 V servo	Same buck, separate terminals	MG996R	1000 µF; keep wires short

Do not power the servo from the ESP32 3.3 V pin or the USB-C port. Use a 5 V buck rated at least 3 A, a 1000 µF electrolytic near the servo, and a common ground star at the base. The e-stop cuts the 12 V actuator bus; the ESP32 may stay alive on 5 V for logging.

4.2 Pin map

Signal	ESP32-S3 pin	Device	Note
HX711 DT	GPIO1	HX711	ADC1 pin; digital input
HX711 SCK	GPIO2	HX711	Output
Moisture	GPIO3	Capacitive pad	ADC1
I2C SDA	GPIO21	OLED / PCF8574	4.7 k pull-up
I2C SCL	GPIO47	OLED / PCF8574	4.7 k pull-up
STEP	GPIO38	A4988	1/8 microstep
DIR	GPIO39	A4988	Clockwise target
EN	GPIO40	A4988	Active low
Servo	GPIO41	MG996R	PWM 50 Hz
IR beam	GPIO42	Receiver	Digital
Home	GPIO48	Microswitch	Pull-up
Lid reed	GPIO45	Reed	Pull-up
Buzzer	GPIO46	Active buzzer	Optional
LED ring	GPIO43	MOSFET	PWM / on-off
4 × bin IR	PCF8574 P0–P3	IR modules	I2C 0x20

Camera pins are fixed by the ESP32-S3-CAM board; do not reassign them. On the common Freenove ESP32-S3-WROOM CAM board, OV2640 uses GPIO4–18 (SIOD=4, SIOC=5, XCLK=15, D0–D7=11/9/8/10/12/18/17/16, VSYNC=6, HREF=7, PCLK=13). GPIO19/20 are USB; GPIO26–37 are flash/PSRAM; GPIO0 is boot. The table above deliberately uses the remaining pins. Confirm against your exact board schematic before soldering.

4.3 Sensor interfaces and wiring

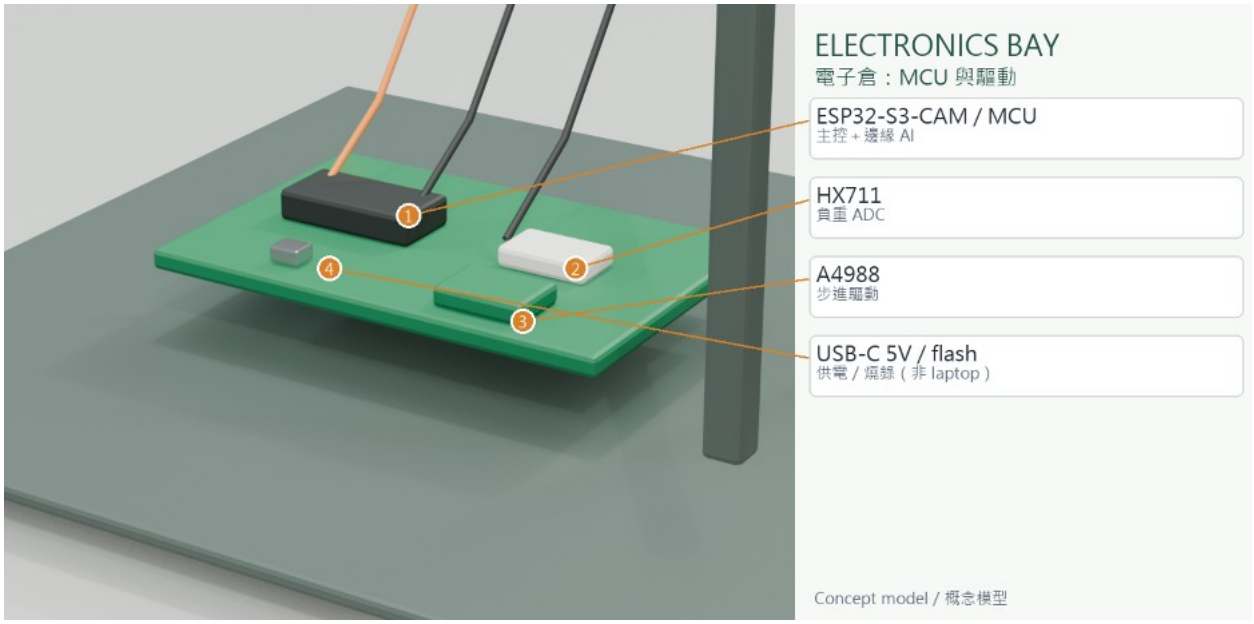


Figure 3. Electronics bay: ESP32-S3, HX711, A4988, USB-C power/flash and wiring.

- HX711: 3.3 V, GND, DT→GPIO1, SCK→GPIO2; shield to GND at HX711 only. Keep away from stepper wires; twisted pair for the load cell.
- Moisture: 3.3 V, GND, AOUT→GPIO3; HDPE cover; calibrate dry/wet; do not use resistive probe in contact with waste.
- IR beam: 5 V, GND, OUT→GPIO14; align across the mouth; 10 k pull-up if open-collector.
- Bin-full IR: 5 V, GND, OUT→PCF8574 P0–P3; mount at 80% height; adjust threshold with a flat target.
- A4988: VMOT 12 V, GND, VDD 3.3 V, STEP/DIR/EN; 100 µF near VMOT; set Vref ≈0.64 V for 0.8 A.
- Servo: 5 V separate terminals, GND, PWM→GPIO13; 1000 µF; do not share the ESP32 3.3 V rail.

4.4 Safety and EMC

- 12 V SELV only; 3 A fuse on the 12 V input; strain relief on all cables; no exposed mains.
- E-stop cuts 12 V to A4988 and servo; software also stops on lid open, drawer open or home switch fault.
- Keep 12 V and camera flex separated; star ground; ferrite on the camera cable if the image is noisy.
- The trapdoor is a pinch point; the lid must be closed before motion. Add a warning label.

4.5 Power budget

Load	Voltage	Typical	Peak	Note
ESP32-S3-CAM	5 V	0.25 A	0.5 A	WiFi + camera
NEMA 17	12 V	0.4 A	0.8 A	Moving only
MG996R	5 V	0.1 A	1.0 A	Stall / start
OLED + sensors	5 V / 3.3 V	0.05 A	0.1 A	—
A4988 logic	3.3 V	0.01 A	0.02 A	—
Total	—	≈0.8 A	≈2.4 A	12 V 3 A / 5 V 3 A

The 12 V 3 A adapter is sized for the stepper and buck, not for simultaneous servo stall. The 5 V buck must supply the ESP32-S3, servo and sensors; use a 3 A part with a heatsink. Add a 3 A fuse and a 1000 µF bulk capacitor at the servo.

4.6 Grounding, noise and EMC

- Single-point star ground at the base; separate high-current return (stepper/servo) from signal return (HX711/camera).
- Twisted pair for the load cell; shield connected at HX711 only; keep HX711 away from the stepper and servo wires.
- Camera flex away from motor phases; add a ferrite if the image has rolling bands.
- Use ferrules on stranded wires; do not tin wires into screw terminals; label both ends.
- ESD: touch grounded frame before handling the S3-CAM; keep spare boards in anti-static bags.

4.7 Connector pinouts

Connector	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6
J1 12 V in	12 V	GND	—	—	—	—
J2 5 V out	5 V	GND	—	—	—	—
J3 HX711	3.3 V	GND	DT	SCK	—	—
J4 A4988	VMOT	GND	VDD	STEP	DIR	EN
J5 servo	5 V	GND	PWM	—	—	—
J6 platform	5 V	GND	Servo PWM	Moisture	Door sw	—
J7 lid	3.3 V	GND	Camera	IR	Reed	LED
J8 bin IR	5 V	GND	P0	P1	P2	P3

Connector numbers are suggestions; use keyed connectors so J3 cannot be plugged into J4. Verify pin 1 orientation on the physical board; the ESP32-S3-CAM camera connector is a flex cable, not a JST header.

5. Firmware, AI and sensor fusion

5.1 Firmware architecture and state machine

Use ESP-IDF or Arduino/PlatformIO with a non-blocking loop. The main loop runs at 100 Hz; no delay() longer than 1 ms. A state machine handles BOOT, HOME, IDLE, ITEM, WEIGH, CAPTURE, CLASSIFY, ROTATE, DUMP, CLOSE, REPORT and FAULT.

State	Entry condition	Action	Exit
BOOT	Power on	Self-test, read switches	HOME
HOME	Boot / fault clear	Home spout clockwise; close door	IDLE
IDLE	Lid closed, drawer closed	LED blue; wait for lid open	ITEM
ITEM	Lid open + IR beam	Wait for stable weight	WEIGH
WEIGH	Item present	Tare, average 20 samples	CAPTURE
CAPTURE	Weight valid	3 frames, LED ring on	CLASSIFY
CLASSIFY	Frames ready	MobileNetV2 + fusion	ROTATE
ROTATE	Target bin known	Move spout, verify home	DUMP
DUMP	Spout in position	Open door; wait item clear	CLOSE
CLOSE	Item dropped	Spring close; home spout	REPORT
REPORT	Cycle done	OLED/SD log; LED green	IDLE
FAULT	Any interlock/timeout	Stop, red LED, OLED code	HOME

5.2 On-device vision pipeline

Step	Detail	Budget
Capture	OV2640, 3 frames, LED ring on	≈300 ms
Crop	Centre ROI around platform	≈10 ms
Resize	96×96 or 128×128, RGB888	≈10 ms
Normalize	INT8, per-tensor scale	≈5 ms
Infer	MobileNetV2 0.35/0.5, TFLM	target <1.5 s
Fuse	Camera + weight + moisture + IR	<50 ms

Use transfer learning: freeze the MobileNetV2 backbone, train a 4-class head, then fine-tune the last block. Export INT8 TFLite; convert to a C array with xxd. In TFLM, allocate the arena in PSRAM (2.5–4 MB). Keep the model at 96×96 if latency is high; 128×128 only if it fits and is faster enough.

Dataset: 300–500 images per class, varied lighting, backgrounds and orientations. Split 70/15/15 by session, not by random frame, to avoid leakage. Record a confusion matrix and per-class precision/recall.

5.3 Sensor fusion rules

Condition	Decision	Why
Moisture > wet threshold	WET	Wet waste overrides vision
Metal high + weight > 20 g	METAL	Metal evidence
Camera plastic > 0.55	PLASTIC	Top class
Camera metal > 0.55	METAL	Top class
Camera wet > 0.55	WET	Top class
Confidence < 0.45	OTHER / ask	Do not guess

Why fusion

A single camera struggles with transparent plastic, wet paper and shiny metal. Weight separates a heavy can from a light bottle; moisture catches food waste; IR confirms the item is present. The rules are deliberately simple and testable.

5.4 OLED, WiFi AP and logging

- OLED 128×64: state, class, confidence, weight, bin, error code. Update at 5 Hz; no flicker.
- WiFi AP: SSID SmartBin-xxxx; local page shows last 20 events and bin-full status. Phone is optional; sorting does not depend on it.
- microSD CSV: timestamp, class, confidence, weight, moisture raw, metal, bin, cycle_ms, error. Rotate weekly.
- Privacy: images processed on-board; store only if the user enables it; no cloud/agent.

5.5 Failure modes and recovery

Fault	Detection	Response
Lid open during cycle	Reed switch	Stop motion; OLED E01; wait close
Drawer open	Microswitch	Refuse start; E02
Spout home missed	Home switch	Retry once; then E03
Load cell timeout	HX711 ready	E04; do not drop
Camera init fail	ESP32 camera	E05; ask user / manual
SD write fail	mount/write	Warn; continue sorting
Brownout on servo	ADC / reset	E06; check 5 V buck

5.6 Code sketches

These are skeletons, not complete libraries. Keep the main loop non-blocking and put long work in a state.

```
// HX711: 20-sample average, tare each cycle
long readWeight() {
  long sum = 0;
  for (int i = 0; i < 20; i++) sum += scale.read();
  return sum / 20 - tare;
}
```

```
// Stepper: 1200 steps per 90 deg, 1/8 microstep, 3:1
void moveToBin(int bin) {
  int target = bin * 1200;
  int delta = (target - pos + 4800) % 4800; // clockwise only
  digitalWrite(DIR, HIGH);
  for (int i = 0; i < delta; i++) {
    digitalWrite(STEP, HIGH); delayMicroseconds(400);
    digitalWrite(STEP, LOW);  delayMicroseconds(400);
  }
  pos = target;
}
```

```
// TFLM inference skeleton
const tflite::Model* model = tflite::GetModel(model_data);
tflite::MicroInterpreter interpreter(model, resolver, arena, kArenaSize);
interpreter.AllocateTensors();
memcpy(interpreter.input(0)->data.int8, input, kInputBytes);
interpreter.Invoke();
int8_t* out = interpreter.output(0)->data.int8;
// dequantize with output->params.scale / zero_point
```

In the real firmware, replace delayMicroseconds with a timer/acceleration ramp; 400 µs is a starting point. Use a current limit and home switch; never assume the spout is at the target without verifying the home sensor.

5.7 Serial protocol and local API

Message	Direction	Payload	Purpose
EVENT	ESP32 → phone	{t, class, conf, wt, bin, ms}	One sort result
STATUS	ESP32 → phone	{state, bin_full[4], err}	Health / full
CMD	phone → ESP32	{cmd: 'open'|'home'|'cal'}	Service only
CAL	phone → ESP32	{scale, tare}	Load-cell calibration

Use newline-delimited JSON on Serial and WebSocket; no binary protocol. Every command has an ID and a timeout. Sorting is autonomous: if the phone disconnects, the bin still works.

5.8 Dataset and training protocol

11. Collect 300–500 images per class in the real hopper.
12. Include clean, wet, crushed, transparent and label-free items.
13. Record lighting: office, window, LED ring only.
14. Label one class per image; keep an uncertain set.
15. Split 70/15/15 by session; never mix frames from one session.
16. Train head with frozen backbone; fine-tune last block.
17. Export SavedModel; INT8 representative dataset (200 images).
18. Convert TFLite; check input/output shapes and scales.
19. Run TFLM on device; record latency and RAM.
20. Evaluate on 50 physical items; confusion matrix + per-class recall.

5.9 Memory and latency budget

Resource	Budget	Note
Flash model	0.5–1.5 MB	MobileNetV2 0.35/0.5 INT8
Flash app	1.0–1.5 MB	ESP-IDF + TFLM

PSRAM arena	2.5–4 MB	TFLM tensors
Frame buffers	≈100–200 KB	1–2 frames at 128×128
Capture	≈300 ms	3 frames
Preprocess	≈50 ms	crop / resize / int8
Inference	0.8–1.5 s	target
Fusion + UI	<50 ms	rules + OLED

If the model does not fit, reduce input to 96×96 or width to 0.35 before reducing the arena below 2.5 MB; a too-small arena fails at AllocateTensors, not at inference.

6. Cost and bill of materials

BOM HK$866; 15% contingency HK$130; total ≈ HK$996. Excludes phone, tools, shipping and a laptop (not required). Prices are October 2026 Hong Kong/Taobao references.

Item	Qty	Unit	Total
ESP32-S3-CAM (8 MB PSRAM)	1	120	120
5 kg load cell + HX711	1	35	35
NEMA 17 stepper	1	75	75
A4988 driver	1	18	18
MG996R servo	1	35	35
IR break-beam	1	15	15
Capacitive moisture	1	15	15
4 × IR bin-full	4	5	20
OLED 0.96"	1	20	20
microSD 16 GB	1	15	15
Microswitches	3	3	9
12 V 3 A adapter	1	45	45
5 V 3 A buck	1	18	18
Plywood / acrylic	1	160	160
3D-printed brackets	1	45	45
Bins (reused)	4	9	36
Hinges / screws / feet	1	60	60
Wire / connectors	1	45	45
Labels / tape	1	20	20
Misc / spare	1	15	15
Subtotal			866
15% contingency			130
TOTAL			≈996

6.2 Procurement and sourcing

Item	Buy locally / online	Note
ESP32-S3-CAM	Online (Taobao / HK)	Buy 2; check PSRAM
NEMA 17 + A4988	Online	Buy spare A4988
Load cell + HX711	Online	5 kg single-point
Plywood/acrylic	Local shop	Offcuts; cut list
Bins	Reuse / Japan Home	Measure internal
Fasteners	Local hardware	M3/M4 kit

Order long-lead items (S3-CAM, NEMA 17, bearing) in week 1. Buy one spare of the S3-CAM and A4988; a failed board during the demo is the most expensive delay. Keep all invoices for the BOM evidence.

7. Build, calibration and test plan

7.1 Build order

21. Base and frame
22. Bins and drawer
23. Mid-frame and bearing
24. Funnel and spout
25. Platform and trapdoor
26. Electronics and wiring
27. Lid and camera
28. Firmware and calibration

7.2 Calibration

- Load cell: 0/100/500/1000 g; record raw; fit scale; tare each cycle; check centre vs 60 mm offset.
- Moisture: dry air, damp sponge, dry bottle; set threshold between dry and wet with 0.15 hysteresis.
- IR full: flat target at 80% height; adjust potentiometer; confirm each bin.
- Home: 20 homes; ±2 steps; if missed, retry once then FAULT.
- Camera: fixed exposure; focus at platform; check 3 lighting conditions; log confidence.

7.3 Test plan

ID	Test	Pass criterion
UT1	Power rails	12 V/5 V within ±5%; no brownout on servo
UT2	Load cell	±10 g at 0–1 kg; centre/offset recorded
UT3	Camera	3 frames <1.5 s; class confidence stable
UT4	Moisture	dry/wet separation; 5 repeats
UT5	NEMA spout	20 moves to each bin; ±3°
UT6	Trapdoor	20 cycles, no jam; item drops
UT7	IR interlocks	lid/drawer open stops motion
UT8	Bin-full	4 sensors trigger at 80%
ST1	50-item mixed	≥80–85% overall; record confusion matrix
ST2	Jam/cycle	≤1 jam / 50; cycle ≤10 s

Record every test in a CSV/Notebook. A demo video is not evidence; the 50-item table, confusion matrix, cycle time and jam count are.

7.4 16-week detailed schedule

Week	Mechanical	Electrical / software	Milestone
1	Final CAD, BOM, order	Board bring-up plan	Orders placed
2	Cut plywood, base	ESP32-S3-CAM blink / camera	Base + camera

3	Bins, drawer, mid-frame	HX711 / OLED test	Bin section
4	Bearing, spout, NEMA	A4988 / home switch	Spout moves
5	Platform, door, servo	Servo / moisture test	Weigh + drop
6	Lid, camera mount	Camera focus / LED	Full mechanism
7	Wiring, cable routing	State machine skeleton	Manual cycle
8	Labels, drip tray	Load-cell calibration	UT1–UT4
9	Spout alignment	Stepper / servo tests	UT5–UT8
10	Dataset capture	Label / split	Dataset v1
11	—	Train / quantize / TFLM	Model on board
12	—	Fusion rules + OLED	End-to-end cycle
13	50-item test	Log / confusion matrix	System data
14	Jam fixes / cleaning	Error handling	Robustness
15	Report / drawings	Video / demo script	Draft
16	Final assembly	Final demo	Submit

7.5 Tools, jigs and consumables

Tool / jig	Use	Note
Jigsaw / hole saw	Plywood / acrylic	Masking tape to reduce chip-out
Drill + bits	Pilot holes	3/4/6 mm
Soldering iron	Headers / connectors	Lead-free solder
Multimeter	Continuity / 5 V / 12 V	Check before power
Scale / weights	Load-cell calibration	0/100/500/1000 g
Bearing jig	Collar alignment	3D-print a centring ring

8. Risks, limitations and future work

Risk	Mitigation
S3 inference too slow	96×96, 0.35 model, 3 frames max
Model accuracy low	Add data, augment, fusion fallback
Spout torque/friction	Lighten spout, 4:1 belt, bearing
Trapdoor jam	Spring assist, chamfer, current limit
Load cell drift	Tare each cycle, shield, warm-up
Wet waste hygiene	Removable tray, HDPE cover, clean
Cost overrun	15% buffer, reuse bins, local buy

- Not full MobileNetV2: 0.35/0.5 at 96–128 px; report the exact model and measured accuracy.
- Not a manufacturing CAD model; no tolerance stack, DFM or safety certification.
- Single load cell: centre the item; off-centre error must be measured.
- IR bin-full is a switch, not a continuous level sensor.
- Metal is inferred from vision + weight; the optional coil is not a certified detector.
- Future: 4× load cells, ToF level, local model update, quieter belt, sealed electronics.

8.3 Mapping to final-year assessment

Criterion	Evidence in this package
Problem and objectives	Section 1; requirements table
Design and analysis	Sections 2–4; torque/clearance checks
Implementation	Sections 5–7; code, pin map, build order
Testing and evaluation	UT1–ST2; 50-item CSV template
Professional practice	Safety, privacy, BOM, maintenance
Communication	A3 drawings, renders, video, report
Limitations	Section 8 + model card

8.4 Future work

- Four load cells for true off-centre weighing.
- ToF or ultrasonic continuous bin level.
- Quieter belt / closed-loop spout encoder.
- On-device incremental learning or local model update.
- Sealed electronics and a removable wet-waste cassette.
- Production DFM, tolerance stack and safety certification.
- Optional local MQTT dashboard; no cloud by default.

Appendix A. Detailed BOM and procurement

ID	Item	Specification	Qty	Unit	Priority
E-01	ESP32-S3-CAM	8 MB PSRAM, 16 MB flash, OV2640, WiFi	1	120	Core
E-02	5 kg load cell	Single-point, 20 kg ultimate	1	25	Core
E-03	HX711	24-bit, 10/80 Hz	1	10	Core
E-04	NEMA 17	42BYGH, 0.45 N·m, 1.7 A	1	75	Core
E-05	A4988	With heatsink	1	18	Core
E-06	MG996R	Metal gear, 10 kg-cm	1	35	Core
E-07	IR break-beam	940 nm, 5 V	1	15	Core
E-08	Moisture pad	Capacitive, 3.3 V	1	15	Core
E-09	IR bin-full	FC-51 or equivalent	4	5	Core
E-10	OLED	0.96 in, 128x64, I2C	1	20	Core
E-11	microSD	16 GB, Class 10	1	15	Core
E-12	Microswitches	Lid, home, drawer	3	3	Core
P-01	12 V 3 A adapter	CE/BS, 5.5×2.1 mm	1	45	Core
P-02	5 V 3 A buck	12 V → 5 V, terminals	1	18	Core
M-01	Plywood/acrylic	12 mm, 420×900-ish	1	160	Core
M-02	3D-printed parts	PETG, ~0.5 kg	1	45	Core
M-03	Bins	170×170×300 mm, reused	4	9	Core
M-04	Hinges/fasteners	Piano hinge, M3/M4, feet	1	60	Core
M-05	Wire/connectors	22–24 AWG, JST/Dupont	1	45	Core
M-06	Labels/tape	Bilingual, colour	1	20	Core
X-01	Metal coil kit	Optional, 3–5 V	1	30	Optional
X-02	Spare servo	MG996R or MG90S	1	35	Optional

Appendix B. Connector and cable schedule

From	To	Cable	Connector	Note
Base	Lid	Camera flex + 4-core	Grommet	Service loop
Base	Platform	6-core	JST-XH	Servo, moisture, switch
Base	NEMA 17	4-core shielded	JST-XH	Keep away from HX711
Base	Home switch	2-core	JST-XH	Pull-up
Base	4 × bin IR	4-core + 5 V/GND	JST-XH	PCF8574
Base	OLED	4-core	JST-XH	I2C

Appendix C. Pseudocode and fusion logic

Main loop (non-blocking), simplified:

```
loop():
  read switches, HX711, moisture, IR
  update state_machine()
  if state == CAPTURE: capture_3_frames()
  if state == CLASSIFY: p = tflm_infer(frame); fuse(p, weight, moisture, metal)
  if state == ROTATE: move_spout(target); verify_home()
  if state == DUMP: open_door(); wait_item_clear(); close_door()
  log_csv(); update_oled()
```

Fusion order: moisture override → metal override → camera top class if confidence ≥0.55 → fallback OTHER/ask. Never drop if weight is invalid or the target bin is full.

Appendix D. 50-item test record

#	Item	True	Pred.	Conf.	Weight	Moist.	Bin	Time	Jam	OK
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										

Appendix E. References and datasheets

- Espressif, ESP32-S3 Technical Reference Manual.
- Espressif, ESP32-S3-CAM board schematic / pinout (board vendor).
- TensorFlow Lite Micro, MicroInterpreter API and arena sizing.
- Sandler et al., MobileNetV2: Inverted Residuals and Linear Bottlenecks, CVPR 2018.
- Allegro, A4988 DMOS Microstepping Driver datasheet.
- Avago/Broadcom, HX711 24-bit ADC datasheet.
- TowerPro, MG996R servo specifications.
- IEC 60335 / local SELV guidance for low-voltage prototypes (reference only).

Appendix F. Pre-demo checklist

- All screws tight; no loose wire in the spout path.
- 12 V fuse and e-stop tested; 5 V rail 5.0 ±0.25 V.
- Home switch repeatable; 20 cycles without fault.
- Trapdoor spring returns; no finger pinch.
- Camera focused; 3 lighting conditions; OLED readable.
- 50-item CSV exported; confusion matrix printed.
- Bin labels correct; wet bin lined; spare gloves/paper towels.
- Video recorded from the front; audio optional.

Appendix G. Dimension sheet



Figure G1. External dimensions: 420 W × 420 D × 900 H mm.

Feature	Dimension
Overall	420 × 420 × 900 mm
Bin	170 × 170 × 300 mm each
Bin spacing	30 mm between bins
Platform	220 × 180 mm; door 180 × 140 mm
Funnel	220 × 200 → 75 mm; 90 mm high
Spout	75 mm elbow; 100 mm outlet radius
Tray-to-lid	225 mm clear
Loading height	≈ 900 mm

Appendix H. Model card and training record

Field	Value
Backbone	MobileNetV2 0.35 / 0.5
Input	96×96 or 128×128 RGB
Output	4 classes: plastic, metal, wet, other
Quantization	INT8, per-tensor
Dataset	300–500 images/class; 70/15/15 by session
Augmentation	flip, brightness, contrast, rotation ±15°
Arena	2.5–4 MB PSRAM
Latency target	<1.5 s per 3-frame cycle
Metrics	accuracy, per-class precision/recall, confusion matrix

Training recipe: (1) collect and label; (2) split by session; (3) train head with frozen backbone; (4) fine-tune last block; (5) export SavedModel; (6) INT8 representative dataset; (7) convert TFLite; (8) xxd to C array; (9) run TFLM on device; (10) record confusion matrix and latency.

Appendix I. Safety risk assessment

Hazard	Risk	Control
Pinch at trapdoor	Finger caught	Lid closed before motion; guard; warning
Spout rotation	Hand in hopper	IR beam; lid interlock; low torque
Electrical	12 V short	Fuse; SELV; strain relief; e-stop
Wet waste	Slip / hygiene	Drip tray; gloves; clean; liner
Lid drop	Finger / head	Gas strut or hinge friction; warning
Sharp edges	Cuts	Sand edges; edge trim
Noise	Stepper resonance	Microstep; belt tension; dampers

Appendix J. 12-week build schedule

Week	Task	Deliverable
1	Finalise design, order long-lead parts	BOM + orders
2	Cut frame, assemble base	Base frame
3	Bins, drawer, mid-frame	Bin section
4	Bearing, spout, NEMA 17	Spout assembly
5	Platform, trapdoor, servo	Weigh assembly
6	Lid, camera, wiring	Electronics
7	Firmware state machine	Manual cycle
8	Calibration, sensor tests	UT records
9	Dataset, training, quantization	Model + metrics
10	50-item test, jam fixes	System data
11	Report, drawings, video	Draft deliverables
12	Final demo, cleanup	Final package

Appendix K. Unit-test record template

Test	Date	Setup	Result	Pass?	Notes
UT1 power					
UT2 load					
UT3 camera					
UT4 moisture					
UT5 spout					
UT6 door					
UT7 IR					
UT8 full					
UT9 log					
UT10 fault					

Appendix L. Fault and repair log

Date	Symptom	Cause	Fix	Time

Appendix M. Glossary

Term	Meaning
Standalone	Runs without a laptop or cloud
TFLM	TensorFlow Lite Micro
INT8	8-bit quantized model
Arena	TFLM working memory
HX711	24-bit load-cell ADC
GT2	2 mm pitch timing belt
SELV	Safety extra-low voltage
DFM	Design for manufacture
Fusion	Combine camera + weight + moisture + IR

Appendix N. Wiring schedule

From	To	Signal	Wire	Note
12 V in	Fuse	12 V	20 AWG	3 A
Fuse	E-stop	12 V	20 AWG	Switch both?
E-stop	A4988 VMOT	12 V	20 AWG	100 µF
E-stop	5 V buck	12 V	20 AWG	—
5 V buck	ESP32-S3 5V	5 V	22 AWG	3 A
5 V buck	Servo	5 V	20 AWG	1000 µF
ESP32	HX711	3.3 V/GND/DT/SCK	24 AWG	Twisted
ESP32	A4988	STEP/DIR/EN	24 AWG	Short
ESP32	OLED	3.3 V/GND/SDA/SCL	24 AWG	I2C
ESP32	PCF8574	3.3 V/GND/SDA/SCL	24 AWG	0x20
PCF8574	4 × bin IR	P0–P3	24 AWG	5 V/GND
ESP32	Lid reed / home	GPIO/GND	24 AWG	Pull-up
ESP32	IR beam	5 V/GND/OUT	24 AWG	Align
ESP32	Moisture	3.3 V/GND/AOUT	24 AWG	ADC1
ESP32	Servo	PWM	24 AWG	Separate 5 V

Appendix O. Spares and consumables

Item	Qty	Why
ESP32-S3-CAM	1 spare	Most likely failure
A4988	1 spare	Cheap; easy to kill
MG996R	1 spare	Gear wear
JST/Dupont	1 kit	Connectors
Heat-shrink	1 pack	Strain relief
Gloves / wipes	1 box	Wet waste
Bin liners	1 roll	Hygiene

Appendix P. 50-item result summary

Metric	Target	Measured	Pass?
Overall accuracy	≥80–85%		
Plastic recall	≥80%		
Metal recall	≥80%		
Wet recall	≥80%		
Other recall	≥75%		
Average cycle	≤10 s		
Jams	≤1 / 50		
Wrong-bin drops	≤10%		